
EXERCISE

Lecture 3: Optimization Recap

AI508 — Algorithmic Game Theory

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Definition: Convex Set

A set $\mathcal{X} \subseteq \mathbb{R}^d$ is *convex* if, for every $x, y \in \mathcal{X}$ and every $\lambda \in [0, 1]$,

$$\lambda x + (1 - \lambda)y \in \mathcal{X}$$

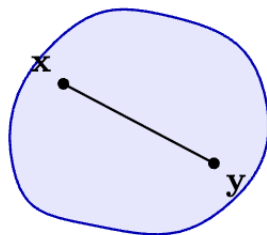
The point $\lambda x + (1 - \lambda)y$ is a *convex combination* of x and y . More generally if $\lambda_i \geq 0$ and $\sum_{i=1}^m \lambda_i = 1$ then,

$$\sum_{i=1}^m \lambda_i x_i$$

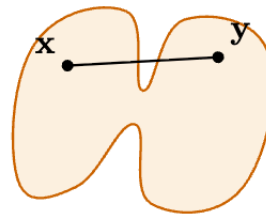
is a convex combination of the point $\{x_1, x_2, \dots, x_n\}$

Important examples of convex sets include affine subspaces, half-spaces, Euclidean balls, boxes, polytopes, and the probability simplex

$$\Delta_d \stackrel{\text{def}}{=} \left\{ x \in \mathbb{R}^d : x_i \geq 0 \ \forall i, \sum_{i=1}^d x_i = 1 \right\}$$



Convex



Not convex

Figure 1: In a convex set, every line segment between two feasible points remains feasible

1 Convex Functions

To define convexity of a function, we assume that its domain is convex.

Definition: Convex function

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be convex. A function $f : \mathcal{X} \rightarrow \mathbb{R}$ is *convex* if, for all $x, y \in \mathcal{X}$ and $\lambda \in [0, 1]$,

$$\lambda x + (1 - \lambda)y \leq \lambda f(x) + (1 - \lambda)f(y).$$

Geometrically, the value of a convex function on a line segment lies below the straight line joining the function values at the endpoints.

Note that a function f is *concave* if $-f$ is convex. It is *strictly convex* if the definition inequality is strict whenever $x \neq y$ and $\lambda \in (0, 1)$.

Example: Convex functions

The functions

$$f(x) = x^2, \quad f(x) = \|x\|_2^2, \quad f(x) = e^x$$

are convex. A linear function $f(x)$ is both convex and concave

2 Jensen's inequality

The two-point definition immediately extends to any finite convex combination.

Theorem: Jensen's inequality

Let $f : \mathcal{X} \rightarrow \mathbb{R}$ be *convex*. Let $x_1, \dots, x_m \in \mathcal{X}$ and let $\lambda_i \geq 0$ satisfy $\sum_{i=1}^m \lambda_i = 1$. Then

$$f\left(\sum_{i=1}^m \lambda_i x_i\right) \leq \sum_{i=1}^m \lambda_i f(x_i).$$

Corollary: Jensen's inequality on a discrete random variable

If X is a finitely supported random variable and f is convex, then

$$f(\mathbb{E}[X]) \leq \mathbb{E}(f(x)).$$

3 First-order characterization of convexity

For differentiable functions, convexity can be characterized by tangent hyperplanes.

Theorem: First-order characterization

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be convex and let $f : \mathcal{X} \rightarrow \mathbb{R}$ be differentiable. Then f is convex if and only if, for all $x, y \in \mathcal{X}$,

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$$

So for a differentiable convex function, the tangent hyperplane at any point is a **global lower bound** on the function.

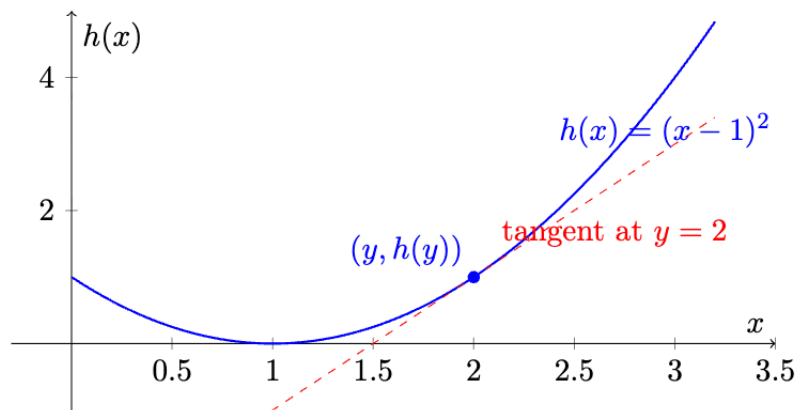


Figure 2: For a differentiable convex function, every tangent line lies below the graph.

4 Strong convexity and smoothness

Definition: Strong convexity

A differentiable function $f : \mathcal{X} \rightarrow \mathbb{R}$ is α -strongly convex for $\alpha > 0$, if for all $x, y \in \mathcal{X}$,

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{\alpha}{2} \|x - y\|_2^2$$

Strong convexity says that the function lies not merely above its tangent hyperplane, but above a quadratic bowl around that tangent hyperplane.

A different property (*smoothness*) controls how quickly the gradient can change.

Definition: Smoothness

A differentiable function $f : \mathcal{X} \rightarrow \mathbb{R}$ is L -smooth if

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L \|x - y\|_2$$

more intuitively:

$$\frac{\|\nabla f(x) - \nabla f(y)\|_2}{\|x - y\|_2} \leq L$$

for all $x, y \in \mathcal{X}$

For an L -smooth function one has the useful upper quadratic bound

$$f(x) \leq f(y) + \langle \nabla f(y), x - y \rangle + \frac{L}{2} \|x - y\|_2^2$$

Thus, strong convexity gives a quadratic lower bound and smoothness gives a quadratic upper bound.

5 Second-order conditions

When a function is twice differentiable, its *Hessian* provides a convenient way to test convexity.

Theorem: Second-order characterization

Suppose f is *twice differentiable* on an open convex set containing $\mathcal{X}$. Then

$$f \text{ is convex} \iff \nabla^2 f(x) \succeq 0 \quad \forall x \in \mathcal{X}$$

Here $\nabla^2 f(x) \succeq 0$ means that

$$v^\top \nabla^2 f(x) v \geq 0 \quad \forall v \in \mathbb{R}^d$$

Hessian is *positive semidefinite*.

The same idea can be applied for α -strongly convex

$$\nabla^2 f(x) \succeq \alpha I \implies f \text{ is } \alpha\text{-strongly convex}$$

and, for twice differentiable convex f

$$0 \preceq \nabla^2 f(x) \preceq LI \iff f \text{ is } L\text{-smooth}$$

Example:

For

$$f(x) = \frac{1}{2}x^\top Qx + c^\top x + r,$$

where $Q = Q^\top$, we have

$$\nabla f(x) = Qx + c, \quad \nabla^2 f(x) = Q$$

Hence f is convex exactly when, $Q \succeq 0$, and it is α -strongly convex when $Q \succeq \alpha I$.

6 Usefulness of convexity: local and global minima

The central structural fact is that convex functions does not have “bad” local minima.

Theorem: Local minima are global under convexity

Let $\mathcal{X}$ be convex and let $f : \mathcal{X} \rightarrow \mathbb{R}$ be convex. Every local minimizer of f over $\mathcal{X}$ is a **global** minimizer

Strict convexity gives *uniqueness*

Theorem: Uniqueness under strict convexity

If f is strictly convex on a convex set $\mathcal{X}$, then f has at most one global minimizer. Consequently, if a minimizer exists, it is unique.

7 First-order optimality over a convex set

For constrained convex optimization, the gradient need not be zero at an optimum: the boundary may prevent us from moving in the direction of $-\nabla f$. The correct condition is a variational inequality

Theorem: First-order optimality condition

Let $\mathcal{X}$ be convex and $f : \mathcal{X} \rightarrow \mathbb{R}$ be differentiable and convex. A point $x^* \in \mathcal{X}$ is a *Global minimizer* if and only if

$$\nabla f(x^*)^\top (x - x^*) \geq 0 \quad \forall x \in \mathcal{X}.$$

If x^* is an interior point of $\mathcal{X}$, then we may move a small amount in both directions along every coordinate. The condition above then reduces to

$$\nabla f(x^*) = 0$$

8 Gradient descent and projected gradient descent

For an unconstrained differentiable objective, gradient descent repeatedly moves opposite to the gradient. For smooth convex objectives, appropriate step sizes yield convergence to a global minimizer.

$$x_{k+1} = x_k - \alpha_k \nabla f(x_k).$$

The stepsize $\alpha_k > 0$ controls how far we move.

When the feasible set is a proper subset $\mathcal{X} \subset \mathbb{R}^d$, a gradient step may leave the feasible region. Projected gradient descent corrects this by projecting back:

$$x_{k+1} = \Pi_{\mathcal{X}}(x_k - \alpha_k \nabla f(x_k)),$$

where

$$\Pi_{\mathcal{X}}(z) \stackrel{\text{def}}{=} \arg \min_{x \in \mathcal{X}} \|x - z\|_2$$

For closed convex $\mathcal{X}$, the Euclidean projection exists and is unique.

9 Constrained optimization: the Lagrangian and KKT conditions

Consider the constrained marginalization problem

$$\begin{aligned} \min_{x \in \mathcal{X}} & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0, \quad i = 1, \dots, n \\ & h_j(x) = 0, \quad j = 1, \dots, m \end{aligned} \tag{1}$$

We associate a nonnegative multiplier λ_i with each inequality constraint and an unrestricted multiplier μ_i with each equality constraint.

Definition: Lagrangian

The *Lagrangian* of the problem above [Equation 1].

$$\mathcal{L}(x, \lambda, \mu) = f(x) + \sum_{i=1}^n \lambda_i g_i(x) + \sum_{j=1}^m \mu_j h_j(x)$$

At an optimum, the Lagrange multipliers measure how the active constraints balance the gradient of the objective.

Theorem: Karush-Kuhn-Tucker conditions

Suppose f and $g_1, \dots, g_m$ are differentiable convex functions and the equality constraints h_j are affine. Under a standard constraint qualification such as Slater's condition, a feasible point x^* is optimal if and only if there exists multipliers λ^*, μ^* satisfying:

(i) **Primal feasibility:**

$$g_i(x^*) \leq 0, \quad h_j(x^*) = 0.$$

(ii) **Dual feasibility:**

$$\lambda_i^* \geq 0.$$

(iii) **Stationarity:**

$$\nabla f(x^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(x^*) + \sum_{j=1}^r \mu_j^* \nabla h_j(x^*) = 0$$

(iv) **Complementary slackness:**

$$\lambda_i^* g_i(x^*) = 0 \quad \forall i$$

Slater's condition means, roughly, that there exists a feasible point satisfying all nonlinear inequality constraints strictly. It is a convenient condition that rules out degenerate pathologies.

Example: A one-dimensional KKT calculation

Consider

$$\begin{aligned} \min_x \quad & x^2 \\ \text{s.t.} \quad & x \geq 1. \end{aligned}$$

Write the constraint as

$$g(x) = 1 - x \leq 0$$

The Lagrangian is

$$\mathcal{L}(x, \lambda)_{\lambda \geq 0} = x^2 + \lambda(1 - x)$$

The KKT conditions are

$$x \geq 1, \quad \lambda \geq 0, \quad 2x - \lambda = 0, \quad \lambda(1 - x) = 0.$$

If $\lambda = 0$, stationarity gives $x = 0$, which is infeasible. Hence the constraint must be active:

$$x^* = 1.$$

Stationarity then gives

$$\lambda^* = 2.$$

Thus the constrained optimum is $x^* = 1$, even though the unconstrained minimizer of x^2 is 0

10 Linear programming

A Linear program (LP) optimizes a linear objective over a feasible set defined by linear constraints. LPs are among the most important optimization problems in algorithms, economics, operations research and game theory.

linear programming is solvable in polynomial time

10.1 Primal Linear Programming

Example: Primal LP

One convenient primal form is

$$\begin{aligned} \max_{x \in \mathbb{R}^n} \quad & c^\top x \quad (\text{primal problem}) \\ \text{s.t.} \quad & Ax \leq b, \\ & x \geq 0 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$ and $c \in \mathbb{R}^n$.

The feasible region of an LP is a polyhedron. When bounded, it is a polytope. Many equivalent LP forms are used in the literature. Equalities can be represented by two inequalities, unrestricted variable can be written as differences of nonnegative variables, and minimization can be converted to maximization by negating the objective.

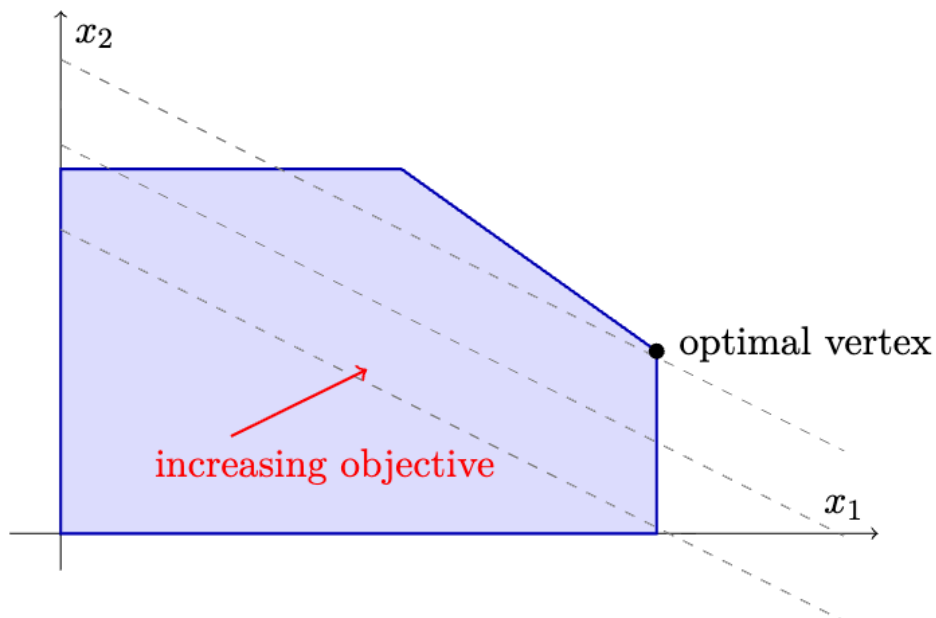


Figure 3: An LP optimizes a linear objective over a polyhedral feasible region. When an optimum exists on a bounded polytope, at least one optimal solution is a vertex.

10.2 Dual Linear Programming

Example: Dual LP

Associated with the primal problem is the *dual* problem

$$\begin{aligned} \max_{x \in \mathbb{R}^n} & b^\top y \quad (\text{dual problem}) \\ \text{s.t.} & A^\top y \geq c, \\ & y \geq 0 \end{aligned}$$

The primal has n variables and m inequality constraints; the dual has m variables and n inequality constraints. The dual variables can be interpreted as prices or shadow values attached to the primal constraints.

10.3 Weak duality

Theorem: Weak duality

For every *primal-feasible* x and *dual-feasible* y ,

$$c^\top x \leq b^\top y$$

10.4 Strong duality

The remarkable fact for LPs is that, under the usual feasibility/boundedness assumptions, the best such upper bound is exact

Theorem: Strong duality for LP

If the primal LP has a finite optimal value, then the dual has an optimal solution and

$$\max_{x \text{ feasible}} c^\top x = \min_{y \text{ feasible}} b^\top y$$

Equivalently, whenever *both* sides admit optimal solutions x^*, y^*

$$c^\top x^* = b^\top y^*$$

We will use strong duality as a standard theorem. There are several proofs, for example through the separating-hyperplane theorem, *Farkas' lemma*, or the **simplex method**.

10.5 Complementary slackness

Strong duality gives an especially useful way to recognize optimal primal–dual pairs.

Theorem: Complementary slackness

Let x^* be primal feasible and y^* dual feasible. They are both optimal if and only if

$$y_i^*(b_i - (Ax^*)_i) = 0 \quad i = 1, \dots, m$$

and

$$x_j^*((A^T y^*)_j - c_j) = 0 \quad j = 1, \dots, n$$

Complementary slackness says, informally:

- dual variable can be positive only when the corresponding primal constraint is tight;
- primal variable can be positive only when the corresponding dual constraint is tight.

Example: Primal, dual and complementary slackness

Consider

$$\begin{aligned} \max_{x_1, x_2 \geq 0} \quad & 3x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + x_2 \leq 4, \\ & x_1 \leq 2, \\ & x_2 \leq 3. \end{aligned}$$

The dual is

$$\begin{aligned} \max_{y_1, y_2 \geq 0} \quad & 4y_1 + 2y_2 + 3y_3 \\ \text{s.t.} \quad & y_1 + y_2 \leq 3, \\ & y_1 \leq 2. \end{aligned}$$

Take

$$x^* = (2, 2), \quad y^* = (2, 1, 0).$$

The primal value is

$$3(2) + 2(2) = 10$$

and the dual value is $4(2) + 2(1) + 3(0) = 10$

Thus weak duality already certifies that both solutions are optimal.

Complementary slackness is also visible

- the first and second primal constraints are tight, and their dual variables y_1^*, y_2^* may be positive
- the third primal constraint is slack ($x_2^* = 2 < 3$), and indeed $y_3^* = 0$;
- both primal variables are positive, and both dual constraints hold with equality.

11 Exercises

2.34 Consider the optimization problem

$$\begin{aligned} & \max_{x_1, x_2, v} v \\ & s.t \quad v \leq 2x_1 \\ & \quad \quad v \leq x_2 \\ & \quad x_1 + x_2 = 1 \\ & \quad x_1, x_2 \geq 0 \end{aligned}$$

1. Solve the linear program and find the optimal values of x_1, x_2 and v
2. Explain why at the optimum, the two constraints

$$v \leq 2x_1 \quad \text{and} \quad v \leq x_2$$

must both be tight

Hint: If one of the two quantities $2x_1$ or x_1 is strictly larger other the other, can you change x_1, x_2 slightly to increase the other.

1. Maximizing both x_1 and x_2 gives,

$$2x_1 = x_2$$

Therefore

$$v = \frac{2}{3}, \quad x_1 = \frac{1}{3}, \quad x_2 = \frac{2}{3}$$

2. Since $v = 2x_1 = x_2$. This is tight

2.35 Consider the regularized optimization problem

$$\max_{x \in \Delta_2} \left\{ g^\top x - \frac{1}{2\eta} \|x\|_2^2 \right\},$$

where

$$g = (g_1, g_2) \in \mathbb{R}^2, \quad \eta > 0,$$

and

$$\Delta_2 = \{x \in \mathbb{R}^2 : x_1, x_2 \geq 0, x_1 + x_2 = 1\}$$

1. Write the lagrangian for this optimization problem
2. Assuming that the optimal solution satisfies

$$x_1^* > 0, \quad x_2^* > 0,$$

write down the first-order optimality conditions.

3. Use these conditions to derive a relation between $x_1^* - x_2^*$ and $g_1 - g_2$.

Hint. Use a lagrange multiplier for the constraint

$$x_1 + x_2 = 1$$

Under the interior assumption, the nonnegativity constraints are inactive

1. Given that $f(x) = g_1 x_1 \cdot g_2 x_2 - \frac{1}{2\eta} \sqrt{x_1^2 + x_2^2}^2$

The constrains are:

$$x_1 + x_2 - 1 = 0, \quad x_1, x_2 \geq 0$$

Therefore

$$\mathcal{L}(x, \lambda, \mu \geq 0) = f(x) - \mu_1 x_1 - \mu_2 x_2 + \lambda(x_1 + x_2 - 1)$$

2. Input x^* as x in the KKT conditions

2.1. **Primal feasibility:** $x_1^* + x_2^* - 1 = 0, \quad x_i^*, x_2^* > 0.$

2.2. **Dual feasibility:** $\lambda_i^* \geq 0.$

2.3. **Stationarity:** $\nabla \mathcal{L} = \begin{pmatrix} g_1 \\ g_2 \end{pmatrix} - \frac{1}{\eta} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 0$

2.4. **Complementary slackness:** $\lambda_i^* g_i(x^*) = 0 \quad \forall i$

3. Given stationarity and Complementary slackness

$$\nabla_{x_1} \mathcal{L} = \nabla_{x_2} \mathcal{L} \implies g_1 - \frac{x_1}{\eta} = g_2 - \frac{x_2}{\eta}$$

$$g_1 - g_2 = \frac{x_1 - x_2}{\eta}$$